Intermediate Elective

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Set Up

Packages

```
# general use
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
# file organization
library(here)
```

```
here() starts at /Users/zeldareif/Desktop/github/intermediate-elective
```

```
# reading .xlsx files
library(readxl)

# data visualization
```

```r
library(naniar)
library(patchwork)
library(NatParksPalettes)
library(gghighlight)
```

Data

```r
# vegetation
veg <- read_csv(here("data", "veg copy.csv"))
```

```
Rows: 27036 Columns: 17
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (10): monitors, transect_name, vp_axis, site, transect_side, vp_zone, q...
dbl   (6): x1, transect_length, num_quad, transect_distance, percent_cover, ...
date  (1): date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# veg metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

```
Rows: 257 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): year_pool, transect_name
dbl (2): year, num_quad

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```
Rows: 50 Columns: 15
-- Column specification --------------------------------------------------------
Delimiter: ","
```

```
chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")
```

Problems

1. Explain your inspiration

I was inspired by Steven Ponce's "Evolution of MTA Art Materials (1980-2020)" because the figure's characteristics would allow me to show total percent cover of multiple vernal pool vegetation species through time, while highlighting one species per faceted plot.

This figure makes sense for the vernal pool vegetation data set because it allows for a clear comparison of total percent cover between different species with distinct lines.

The plot will be faceted by vegetation species, the x axis will be time by year and the y axis will be total percent cover. Each plotted line will represent a different vernal pool vegetation species, and the species of interest's line will be highlighted with a different color to stand out.

2.

Sketch what your figure would look like following your inspiration.

Insert that sketch (as a photo, screenshot, or otherwise) into your document.

3. Code your figure

```
# create new object from veg
native_spp <- veg |>
  # filter to only include native cover
  filter(str_detect(cover_category, "NATIVE COVER")) |>
  # filter by species
  pull(psoc) |>
```

```r
  # list each unique native species
  unique()

# display 10 random species from data frame
sample(native_spp, 10)
```

```
 [1] "Bromus_diandrus"                       "Quercus_agrifolia"
 [3] "Atriplex_coulteri"                     "Tragopogon_porrifolius"
 [5] "Eleocharis_parvula"                    "Symphyotrichum_subulatum"
 [7] "Malacothris_saxatilis_var_implicata" "Brickellia_Californica"
 [9] "Frankenia_salina"                      "Extriplex_californica"
```

```r
# define function to rename species
categorize_species <- function(species_string) {
  case_when(
    species_string == "Helminthotheca_echioides" ~ "Bristly oxtongue",
    species_string == "Erigeron_canadensis" ~ "Horseweed",
    species_string == "Eryngium_vaseyi" ~ "Coyote-thistle",
    species_string == "Hordeum_brachyantherum_brachy." ~ "Meadow barley",
    species_string == "Festuca_myuros" ~ "Rat-tail fescue",
    species_string == "Bromus_diandrus" ~ "Ripgut brome",
    TRUE ~ "Other")}

# create new object from veg
veg_pc <- veg |>
  # mutate to only include selected species
  mutate(taxon_group = categorize_species(psoc)) |>
  # filter to remove species not selected
  filter(taxon_group != "Other") |>
  # filtered to only include ncos vernal pool transects
  filter(str_detect(transect_name, "VP") &
         site == "ncos") |>
  # grouped by species, transect, and year
  group_by(taxon_group, transect_name, year) |>
  # summarized by calculating total percent cover across all
  # quadrats
  summarise(
    sum_pc = sum(percent_cover, na.rm = TRUE),
    .groups = "drop") |>
  # created column to join with metadata
  unite("year_pool", year, transect_name, remove = TRUE) |>
```

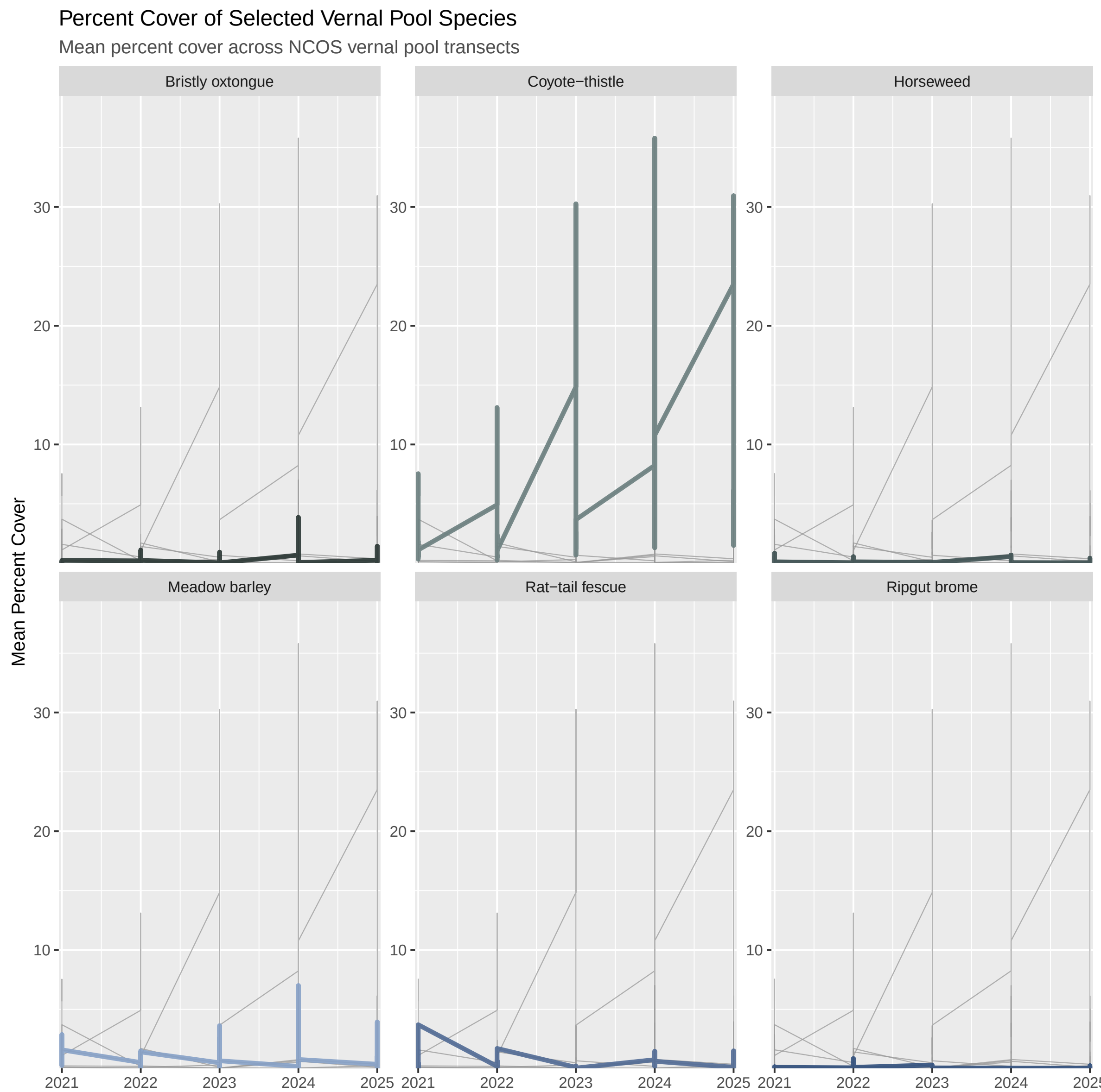
```r
  # joined metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculated mean percent cover
  mutate(mean_pc = sum_pc/num_quad)
```

```r
# get colors from national parks palette
park_colors <- natparks.pals("Yosemite")

# assign colors to species
species_colors <- c(
  "Bristly oxtongue" = park_colors[1],
  "Horseweed" = park_colors[2],
  "Coyote-thistle" = park_colors[3],
  "Meadow barley" = park_colors[4],
  "Rat-tail fescue" = park_colors[5],
  "Ripgut brome" = park_colors[6]
)

### |- titles and caption ----
title_text <- str_glue("Percent Cover of Selected Vernal Pool Species")
subtitle_text <- str_glue("Mean percent cover across NCOS vernal pool transects")
```

```r
  ggplot(data = veg_pc,
    mapping= aes(x = year,
                 y = mean_pc,
                 color = taxon_group)) +

  # Geoms
  geom_line(size = 1.3, alpha = 0.9) +

  gghighlight(
    use_direct_label = FALSE,
    unhighlighted_params = list(
      linewidth = 0.3,
      alpha = 0.75,
      color = "gray60"
    )
  ) +

  # Scales
  scale_color_manual(
    values = species_colors,
```

```r
    guide = "none"
  ) +

  scale_x_continuous(
    expand = expansion(mult = c(0.01, 0.01))
  ) +

  scale_y_continuous(
    expand = expansion(mult = c(0.0, 0.1))
  ) +

  # Labs
  labs(
    title = title_text,
    subtitle = subtitle_text,
    x = NULL,
    y = "Mean Percent Cover"
  ) +

  # Facets (optional but usually helpful for species data)
  facet_wrap(~taxon_group,
             scales = "free_y",
             ncol = 3) +

  # Theme
  theme(
  plot.title = element_text(color = "black"),
  plot.subtitle = element_text(color = "gray30"),
  plot.caption = element_text(color = "gray40"))
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```



4. Describe your process

In 1-3 sentences each, describe:

- your process of using someone else's code to create a visualization
- challenges you encountered as you made your visualization
- how your visualization differs from visualizations you made in class, for assignments, or for a previous elective